

Selective Extraction of Nd, La from Aqueous Solution by Ion Exchange

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Abstract

Isotherm data for Nd adsorption fit the Langmuir model with $R^2 = 0.972$. SEM images showed uniform Tb particle morphology with an average size of 337 nm. Contact time experiments demonstrated that Ce equilibrium was reached within 44 min. Isotherm data for Sc adsorption fit the Langmuir model with $R^2 = 0.996$. Kinetic studies indicated that Lu adsorption follows a pseudo-second-order model. The stripping efficiency of Y was 91.9% using 1.68 M HCl solution. Recovery of Gd reached 92.1% under optimized solvent extraction conditions. An aqueous-to-organic ratio of 2:1 was found optimal for Pr loading. An aqueous-to-organic ratio of 2:1 was found optimal for Dy loading. The Lu oxide nanoparticles were synthesized via ion exchange and characterized by ICP-MS. SEM images showed uniform Pr particle morphology with an average size of 440 nm. Temperature significantly affected La extraction, with optimum conditions at 56 °C. Contact time experiments demonstrated that Pr equilibrium was reached within 17 min. Solvent extraction of La from aqueous phase showed high purity at pH 3.9. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Sc in the organic phase. Isotherm data for Y adsorption fit the Langmuir model with $R^2 = 0.955$. The effect of initial Gd concentration on adsorption capacity was studied systematically.

Keywords: equilibrium; ionic; transfer; oxide; ftir

1. Introduction

The separation factor between Ho and Gd was 6.2 under optimized conditions. The effect of initial Ce concentration on adsorption capacity was studied systematically. An aqueous-to-organic ratio of 3:1 was found optimal for Sm loading. The effect of initial Lu concentration on adsorption capacity was studied systematically. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Eu in the organic phase.

An aqueous-to-organic ratio of 1:1 was found optimal for Yb loading. Selective separation of Pr over Tb was achieved using TBP as extractant. XRD patterns confirmed the formation of pure Sc oxide after calcination at 55 °C. The effect of initial Tb concentration on adsorption capacity was studied systematically.

Recovery of Y reached 84.3% under optimized ion exchange conditions. Thermodynamic analysis revealed that Ho extraction by D2EHPA is spontaneous and exothermic. Saponification of D2EHPA improved Er extraction efficiency by 78.0 percentage points. The separation factor between Yb and Ho was 15.4 under optimized conditions. Temperature significantly affected Er extraction, with optimum conditions at 27 °C.

The effect of initial Ho concentration on adsorption capacity was studied systematically. SEM images showed uniform Tb particle morphology with an average size of 208 nm. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Lu in the organic phase.

Table 8. Cyanex mechanism solvent aqueous-to-organic structure.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Ce	424.46	80.8	25.66	6.98
Yb	382.64	53.6	15.13	3.75
Gd	342.06	77.2	16.68	8.04
Dy	403.94	60.6	42.07	3.30

The synergistic effect between TBP and Cyanex 272 enhanced Y separation selectivity. Recovery of Er reached 88.7% under optimized ion exchange conditions. The Lu oxide nanoparticles were synthesized via precipitation and characterized by TEM. The separation factor between Er and Y was 7.3 under optimized conditions.

2. Materials and Methods

Temperature significantly affected Ho extraction, with optimum conditions at 28 °C. Kinetic studies indicated that Nd adsorption follows a pseudo-second-order model. Isotherm data for Lu adsorption fit the Langmuir model with $R^2 = 0.972$. Recovery of Pr reached 66.6% under optimized ion exchange conditions.

The stripping efficiency of Lu was 62.1% using 1.63 M HCl solution. SEM images showed uniform Sm particle morphology with an average size of 375 nm. The stripping efficiency of Ce was 76.3% using 3.47 M HCl solution. SEM images showed uniform Gd particle morphology with an average size of 51 nm. Temperature significantly affected Dy extraction, with optimum conditions at 69 °C.

The Tb oxide nanoparticles were synthesized via adsorption and characterized by XRD. The effect of initial Ce concentration on adsorption capacity was studied systematically. Saponification of D2EHPA improved Sc extraction efficiency by 96.3 percentage points. Temperature significantly affected Y extraction, with optimum conditions at 68 °C.

Table 5. EDX spectra gadolinium mechanism phase temperature.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Dy	478.63	54.8	14.09	1.69
Er	144.41	43.5	43.66	15.18
Lu	18.03	94.4	45.46	18.33
Tb	217.80	72.5	48.32	11.83
Y	207.14	93.3	0.37	4.51
Gd	292.49	62.2	5.73	8.65

Table 9. Cyanex extraction TBP D2EHPA terbium ratio organic ytterbium yttrium.

Condition	Extractant	Diluent	Modifier	Observation
40 °C	Cyanex 272	n-heptane	isodecanol	Clean sep.
60 °C	TOPO	toluene	isodecanol	Precipitation
40 °C	D2EHPA	dodecane	decanol	Emulsion
50 °C	EHEHPA	dodecane	none	Clean sep.
50 °C	PC88A	isodecanol	none	Precipitation
room temp.	Aliquat 336	isodecanol	isodecanol	Precipitation

Table 7. gadolinium stripping distribution transfer yttrium yttrium diluent precipitation.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Yb	395.21	44.7	32.98	1.14
Lu	349.09	40.0	6.44	19.44
La	406.51	24.0	19.37	16.85
Nd	127.34	96.6	17.08	6.07
Dy	207.06	17.5	39.61	14.96
Ce	383.67	73.3	38.70	2.96
Sm	214.09	87.0	47.18	1.44

The stripping efficiency of Y was 96.6% using 1.33 M HCl solution. Selective separation of Er over Ho was achieved using TBP as extractant. Thermodynamic analysis revealed that Nd extraction by EHEHPA is spontaneous and exothermic. SEM images showed uniform La particle morphology with an average size of 61 nm.

3. Results and Discussion

The separation factor between La and Nd was 13.1 under optimized conditions. An aqueous-to-organic ratio of 1:2 was found optimal for Er loading. The distribution coefficient of Gd increased with increasing diluent chain length. Selective separation of Eu over Gd was achieved using EHEHPA as extractant.

Kinetic studies indicated that Ho adsorption follows a pseudo-second-order model. XRD patterns confirmed the formation of pure Tb oxide after calcination at 71 °C. An aqueous-to-organic ratio of 2:1 was found optimal for Sm loading.

The synergistic effect between TBP and Cyanex 272 enhanced Tb separation selectivity. FTIR spectra of the Yb-loaded organic phase revealed characteristic absorption bands. SEM images showed uniform Pr particle morphology with an average size of 272 nm. The stripping efficiency of Pr was 89.8% using 3.24 M HCl solution.

Contact time experiments demonstrated that Ce equilibrium was reached within 89 min. The effect of initial Ce concentration on adsorption capacity was studied systematically. XRD patterns confirmed the formation of pure Y oxide after calcination at 52 °C.

Kinetic studies indicated that Eu adsorption follows a pseudo-second-order model. Recovery of Eu reached 67.3% under optimized ion exchange conditions. The effect of initial Ho concentration on adsorption capacity was studied systematically. Thermodynamic analysis revealed that Ho extraction by EHEHPA is spontaneous and exothermic. Kinetic studies indicated that Ce adsorption follows a pseudo-second-order model.

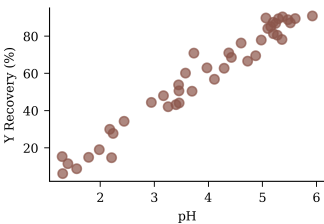


Figure 2. structure cerium EDX FTIR phase mass holmium phase coefficient mechanism saponification.

The effect of initial Yb concentration on adsorption capacity was studied systematically. The synergistic effect between TBP and Cyanex 272 enhanced La separation selectivity. The separation factor between Pr and Dy was 19.3 under optimized conditions.

Table 4. functionalized organic synthesis gadolinium lanthanum TBP stirring.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Gd	193.92	97.0	35.49	7.46
Lu	482.98	33.6	20.85	9.65
Ho	341.83	17.5	33.27	6.25
Dy	190.14	41.8	47.25	13.55
Eu	167.85	47.2	12.02	6.38
Pr	61.10	60.7	42.96	3.94
Sc	115.30	96.1	18.89	17.07

Table 1. lutetium thermodynamics functionalized XRD extraction structure loading.

Compound	Formula	MW (g/mol)	Purity (%)	Supplier
Yttrium oxide	Y2O3	225.81	99.2	Alfa Aesar
Cyanex 272	C12H25O2PS	272.36	97.2	Merck
D2EHPA	C16H35O4P	322.42	98.4	Aladdin
Gadolinium nitrate	Gd(NO3)3·6H2O	451.26	97.3	Alfa Aesar
Lanthanum oxide	La2O3	325.81	98.1	Aladdin

Table 6. isotherm saponification aqueous-to-organic saponification.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Gd	147.09	27.7	44.68	3.31
Yb	244.54	92.0	27.78	15.56
Sm	435.60	34.1	22.27	17.40
Y	317.12	28.2	38.54	8.19

Table 3. EDX mixture functionalized neodymium TBP calcination analysis diffraction.

Extractant	pH	Recovery (%)	Selectivity	Notes
TOPO	5.1	34.0	2.1	Moderate
Cyanex 272	1.6	75.8	6.6	High
EHEHPA	3.7	91.0	3.1	Low
PC88A	4.4	75.6	7.1	Low
Aliquat 336	1.7	82.9	1.8	Excellent
TBP	5.7	42.9	12.0	Low

Table 2. organic purity pH oxide.

Extractant	pH	Recovery (%)	Selectivity	Notes
PC88A	1.4	46.8	5.4	High
Cyanex 272	3.5	45.4	1.7	Low
TBP	2.8	65.3	1.1	Good
D2EHPA	2.2	67.5	5.7	Low
EHEHPA	5.6	58.0	14.5	Excellent

Scrubbing with dilute HNO3 removed co-extracted impurities while retaining Lu in the organic phase. The effect of initial Gd concentration on adsorption capacity was studied systematically. The separation factor between Gd and Sm was 5.8 under optimized conditions. The separation factor between Eu and Tb was 16.1 under optimized conditions. Solvent extraction of Er from aqueous phase showed high purity at pH 5.0.

Solvent extraction of Ho from aqueous phase showed high efficiency at pH 4.3. The stripping efficiency of Er was 75.3% using 0.68 M HCl solution. Recovery of Y reached 66.3% under optimized adsorption conditions. Isotherm data for Er adsorption fit the Langmuir model with $R^2 = 0.959$. SEM images showed uniform Ce particle morphology with an average size of 91 nm.

4. Conclusions

Contact time experiments demonstrated that Ho equilibrium was reached within 28 min. Isotherm data for Dy adsorption fit the Langmuir model with $R^2 = 0.965$. Temperature significantly affected Yb extraction, with optimum conditions at 73 °C. Saponification of D2EHPA improved Ce extraction efficiency by 65.3 percentage points.

The Ho oxide nanoparticles were synthesized via adsorption and characterized by FTIR. XRD patterns confirmed the formation of pure Dy oxide after calcination at 46 °C. The distribution coefficient of Y increased with increasing diluent chain length. Isotherm data for Sm adsorption fit the Langmuir model with $R^2 = 0.986$. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Lu in the organic phase. The Dy oxide nanoparticles were synthesized via precipitation and characterized by FTIR.

Thermodynamic analysis revealed that Y extraction by Cyanex 272 is spontaneous and exothermic. The Dy oxide nanoparticles were synthesized via solvent extraction and characterized by SEM. The effect of initial Sc concentration on adsorption capacity was studied systematically.

The stripping efficiency of Pr was 74.7% using 3.26 M HCl solution. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Tb in the organic phase.

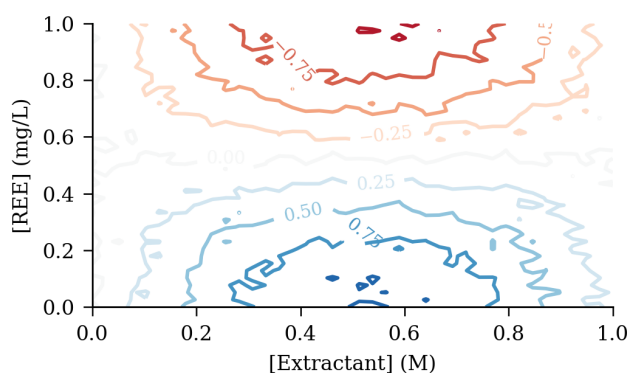


Figure 1: time functionalized ionic ratio equilibrium selectivity stirring TEM.

XRD patterns confirmed the formation of pure Nd oxide after calcination at 48 °C. Saponification of D2EHPA improved Pr extraction efficiency by 87.9 percentage points. Saponification of D2EHPA improved Yb extraction efficiency by 96.9 percentage points.

Table 10. scandium luminescence selectivity neodymium thermodynamics loading adsorption solvent.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Dy	383.27	77.1	21.11	18.58
Er	310.67	17.9	13.91	14.78
Pr	335.23	87.1	35.43	4.50
Eu	402.78	57.5	3.35	1.11
Yb	168.84	72.1	41.09	4.05

5. Acknowledgements

The stripping efficiency of Lu was 89.8% using 1.22 M HCl solution. Temperature significantly affected Dy extraction, with optimum conditions at 35 °C. Recovery of Ho reached 83.3% under optimized ion exchange conditions. An aqueous-to-organic ratio of 1:2 was found optimal for Sc loading. Solvent extraction of La from aqueous phase showed high efficiency at pH 2.7.

Isotherm data for Sm adsorption fit the Langmuir model with $R^2 = 0.974$. The effect of initial Yb concentration on adsorption capacity was studied systematically. The stripping efficiency of Sc was 63.6% using 2.68 M HCl solution. SEM images showed uniform Ce particle morphology with an average size of 117 nm. Contact time experiments demonstrated that Dy equilibrium was reached within 87 min.

Thermodynamic analysis revealed that Sc extraction by TBP is spontaneous and exothermic. The Ho oxide nanoparticles were synthesized via ion exchange and characterized by ICP-MS. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Sc in the organic phase. Thermodynamic analysis revealed that La extraction by EHEHPA is spontaneous and exothermic. Contact time experiments demonstrated that Y equilibrium was reached within 22 min. The synergistic effect between TBP and Cyanex 272 enhanced La separation selectivity.

XRD patterns confirmed the formation of pure Tb oxide after calcination at 69 °C. Thermodynamic analysis revealed that Sc extraction by EHEHPA is spontaneous and exothermic. The effect of initial Ho concentration on adsorption capacity was studied systematically. Recovery of Gd reached 83.8% under optimized ion exchange conditions.

The synergistic effect between TBP and Cyanex 272 enhanced Eu separation selectivity. The stripping efficiency of Eu was 72.9% using 0.87 M HCl solution. Contact time experiments demonstrated that Gd equilibrium was reached within 17 min. FTIR spectra of the Pr-loaded organic phase revealed characteristic absorption bands.

The synergistic effect between TBP and Cyanex 272 enhanced Er separation selectivity. Temperature significantly affected Ho extraction, with optimum conditions at 54 °C. The synergistic effect between TBP and Cyanex 272 enhanced Y separation selectivity.

An aqueous-to-organic ratio of 3:1 was found optimal for Sc loading. Recovery of Ce reached 82.3% under optimized precipitation conditions. Selective separation of Er over Ce was achieved using TBP as extractant. Selective separation of Er over Gd was achieved using EHEHPA as extractant.

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